SmartSpend — FHS Formula, Validation, and Questionnaire Summary

For Ma'am Ellen F. Mangaoang, MIT (Capstone Adviser)

Project: SmartSpend — AI-Assisted Mobile Financial Tracking and Advisory Application
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Purpose: A concise reference for the next adviser meeting (read in ~5 minutes)

A. Literature Basis of the FHS Formula

The Financial Health Score (FHS) is a **Prototype Observed Financial Health Indicator (0–100)**, computed directly from the user's actual logged transaction data — not from self-report surveys. It was adapted from the **CBA–Melbourne Institute Observed Financial Health Indicator** (Comerton-Forde et al., 2018), which derives a 0–100 observed scale from actual bank-record data using equal-weight components. SmartSpend applies this transaction-based approach to a mobile SQLite database instead of bank records.

The FHS is the sum of four components, each worth 25 points:

Component	Formula	Academic Basis
Savings Rate	25 × min(1, savingsRate / 0.20)	Warren & Tyagi (2005) — 50/30/20 rule
Overspend Control	25 × (1 − weightedOverDays / activeDays)	Kahneman & Tversky (1979) — Prospect Theory
Budget Adherence	25 × (onBudgetCategories / totalBudgetCategories)	Financial Health Network FinHealth Score® — Plan pillar
Logging Consistency	25 × (loggedDays / activeDays)	Deci & Ryan (2000) — Self-Determination Theory

Why equal weights (25 each): The OECD/JRC *Handbook on Composite Indicators* (2008) establishes equal weighting as the methodologically correct default when no empirical basis exists for assigning differential weights. Each component contributes equally to the total, consistent with the CBA–Melbourne Institute dual-scale model.

B. Validation

System Usability Scale (SUS): Preliminary results from N = 5 respondents yield a mean SUS of **62.50**, corresponding to **Grade C / "OK"** per the Bangor et al. (2009) interpretation scale. This sits below the established 68-point average benchmark. We acknowledge this as a limitation; the full N = 30 evaluation is ongoing and will be reported in the final manuscript.

Technical validation: OCR precision of **91.2%**, Taglish (Filipino-English) transaction categorization accuracy of **94.6%**, and a **100% database recovery rate** across backup/restore testing.

Formula validation: The formula design follows the CBA–Melbourne Institute dual-scale model (Comerton-Forde et al., 2018, 2020). The prototype status of the indicator is explicitly acknowledged throughout the documentation.

C. Questionnaire Status

A **30-question survey instrument** has been completed and is ready for adviser content validation and signature. It is structured into five sections:

1. **Respondent Profile** (Q1–Q7)
2. **Financial Practices** (Q8–Q13)
3. **Budgeting Challenges** (Q14–Q20)
4. **Expense Tracking** (Q21–Q26)
5. **Supplementary Profile** (Q27–Q30)

Sampling plan: N = 30 respondents — 20 parents aged 35–55 (primary) and 10 young professionals aged 21–35 (secondary) — drawn from San Fernando City, La Union.

The survey is ready for content validation and is **awaiting Ma'am Ellen's signature**.

Full documentation available in: FHS_Formula_For_Maam_Mangaoang.pdf, FHS_Formula_Basis_Table.pdf, FHS_Defense_Guide.pdf